

Choosing a Model from the Evidence

Answer Key

Algebra 1

Quick Answers

- | | |
|-------------------|---------------------------------|
| 1. (b) = 95, — | 7. $m = 3$, $y = 85$ |
| 2. (b) = 50 ft, — | 8. (b) = 84 ft ² , — |
| 3. (b) = 9, — | 9. 10 lb |
| 4. 310 min | 10. (b) = 31, — |
| 5. 0, 4 s | 11. 84 |
| 6. 54 L | 12. (b) = 2, 12, — |
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What is verified

6 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 8, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-CED.A.2 – *problems 3, 4, 5, 6, 7, 9, 11, 12*

HSF-IF.B – *problems 1, 2, 8, 10*

Difficulty 1–5 of 5 across 18 tagged checks.

1. Shirt order: costs 25, 32, 39, 46 for $n = 0, 1, 2, 3$. (a) Which family? (b) Cost of 10 shirts.

(a) First differences: $32 - 25 = 7$, $39 - 32 = 7$, $46 - 39 = 7$. They are constant, so the rule is **linear** — each extra shirt adds the same amount. (A quadratic table would need constant *second* differences; a piecewise table would change its rule partway.)

(b) Constant difference 7 is the rate, and 25 is the value at $n = 0$, so $C(n) = 25 + 7n$. Then $C(10) = 25 + 7(10) = 25 + 70$. 95

2. Cart heights 0, 44, 56, 36 ft at $t = 0, 1, 2, 3$ s. (a) Which family? (b) Height at $t = 2.5$ s.

(a) First differences: 44, 12, -20 — not constant, so not linear. Second differences: $12 - 44 = -32$ and $-20 - 12 = -32$ — constant, so the rule is **quadratic**. Nothing in the table changes rule partway, so it is not piecewise.

(b) Substitute $t = 2.5$ into $h(t) = -16t^2 + 60t$: $-16(6.25) + 60(2.5) = -100 + 150$.

50 ft

3. Garage: \$3 for the first hour, \$2 per hour after. (a) Which family? (b) Charge for 4 hours.

(a) Piecewise. The rule for the first hour (\$3 flat) is not the rule that applies afterwards (\$2 per hour), so no single straight-line rule covers the whole domain — the model needs one piece up to $t = 1$ and a second piece beyond it.

(b) For $t \geq 1$ the charge is $3 + 2(t - 1)$. At $t = 4$: $3 + 2(3) = 3 + 6$.

9

4. Phone plan: \$12 per month plus \$0.05 per minute; bill \$27.50. Find the minutes m .

Solution: the charge grows by the same \$0.05 for every minute, so a linear equation models it:

$$12 + 0.05m = 27.50$$

$$0.05m = 15.50$$

$$m = \frac{15.50}{0.05}$$

$m = 310 \text{ min}$

5. Rocket: $h(t) = -16t^2 + 64t$. Solve $h(t) = 0$.

Solution: the rule is quadratic, so factor rather than divide by t (dividing would lose a root):

$$-16t^2 + 64t = 0$$

$$-16t(t - 4) = 0$$

The zero-product property gives $-16t = 0$ or $t - 4 = 0$.

$t = 0 \text{ s or } t = 4 \text{ s}$

6. Tank: 5 L/min for 8 min, then 2 L/min. Volume after 15 minutes.

Solution: the rate changes at $t = 8$, so the model is piecewise. The first piece finishes with $5 \times 8 = 40$ litres. The second piece then runs for $15 - 8 = 7$ minutes at 2 litres per minute: $40 + 2(7) = 40 + 14$.

54 L

7. Gym A: $40 + 15m$. Gym B: $25 + 20m$. Find the month m and cost y where they agree.

Solution: both fees grow at a constant rate per month, so both are linear. Set the two costs equal:

$$40 + 15m = 25 + 20m$$

$$15 = 5m$$

$$m = 3$$

Then $y = 40 + 15(3) = 85$.

$m = 3 \text{ months, } y = 85$

8. Dog run, 40 ft of fence, $A(w) = w(20 - w)$. (a) Which family? (b) Area at $w = 6$.

(a) Quadratic. Both factors carry the variable: widening the run by one foot lengthens one pair of sides and shortens the other, so w appears twice and the expanded rule $20w - w^2$ is second degree. A linear rule would change the area by the same amount for every extra foot of width, and this one does not.

(b) Substitute $w = 6$: $6(20 - 6) = 6 \times 14$.

$$84 \text{ ft}^2$$

9. Courier: \$6 up to 5 lb, then \$1.20 per extra pound; parcel cost \$12. Find w .

Solution: above 5 pounds the second piece applies, so

$$6 + 1.20(w - 5) = 12$$

$$1.20(w - 5) = 6$$

$$w - 5 = 5$$

$$w = 10 \text{ lb}$$

10. Table $y = 3, 7, 13, 21$ for $x = 1, 2, 3, 4$; the claim “linear, because y goes up every time”. (a) Why is that not enough? (b) Predict y at $x = 5$.

(a) Rising values only rule out a decreasing rule; a quadratic can rise every time too. The test is the differences. First differences: 4, 6, 8 — not constant, so the rule is not linear. Second differences: $6 - 4 = 2$ and $8 - 6 = 2$ — constant, so the table is **quadratic**.

(b) Substitute $x = 5$ into $y = x^2 + x + 1$: $25 + 5 + 1$.

$$31$$

11. Electricity: \$0.11 per kWh for the first 400 kWh, then \$0.16 per kWh. Bill for $k = 650$ kWh.

Solution: the rate changes at 400 kWh, so the model is piecewise and 650 lies in the second piece. First 400 kWh: $0.11 \times 400 = 44$ dollars. The remaining $650 - 400 = 250$ kWh: $0.16 \times 250 = 40$ dollars. Total $44 + 40$.

$$84$$

12. Apples: $R(x) = x(40 - 2x)$, $C(x) = 12x + 48$, both in hundred dollars. (a) Name each family. (b) Solve $R(x) = C(x)$.

(a) R is **quadratic**: the price a crate fetches, $40 - 2x$, falls as the number of crates x rises, so the variable sits in both factors and the expanded rule $40x - 2x^2$ is second degree. C is **linear**: every extra hundred crates adds the same 12, on top of the fixed 48 that is there even at $x = 0$.

(b) Set them equal and collect on one side:

$$x(40 - 2x) - (12x + 48) = 0$$

$$40x - 2x^2 - 12x - 48 = 0$$

$$-2x^2 + 28x - 48 = 0$$

$$x^2 - 14x + 24 = 0$$

$$(x - 2)(x - 12) = 0$$

$$x = 2 \text{ or } x = 12$$